

Hierarchical Data Format Version 5 (HDF5)

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Key Concepts

An HDF5 file is a container for two kinds of objects: *datasets*, which are collections of data in array form, and *groups*, which are folder-like containers holding datasets and other groups. Groups behave like dictionaries, and datasets like NumPy arrays.

Reference: `h5py` Quick Start.

What is HDF5?

HDF5 (Hierarchical Data Format version 5) is a binary format optimized for storing large volumes of numerical data, together with an API (via `h5py` or `PyTables` in Python) that provides:

- *Hierarchical* organization into groups and datasets, similar to a file system;
- the ability to add *attributes* (metadata) to any object;
- high-performance *random access* thanks to chunking and compression.

1. Hierarchical Model

- **Groups** (`Group`): folder-like containers.
- **Datasets** (`Dataset`): multidimensional arrays (`numpy.ndarray`).
- **Attributes** (`attrs`): key-value pairs attached to a group or dataset.

Example structure:

```
/                                % root
|- measurements/                % group
|   |- time                    (dataset)
|   '- temperature             (dataset)
|- parameters/                  % group
|   |- voltage                 (dataset)
|   '- current                 (dataset)
'- metadata                     % group
    |- experiment_date         (attribute)
    '- operator                (attribute)
```

2. Why HDF5 Excels at Organizing Data

HDF5 delivers clear and efficient data management thanks to several strengths:

1. Natural Organization

You arrange your data in a simple tree structure, like folders and files on your computer. For example, `/measurements/temperature` or `/parameters/pressure`. This lets you locate and navigate directly to the information you need.

2. Built-in Metadata

Each group or dataset can carry its own labels (attributes), such as measurement units, creation date, or a description. All relevant information stays together in one place, without external sidecar files.

3. Partial, High-Performance Access

When your arrays are very large, HDF5 can split them into “chunks” and compress them. You can read or write a single chunk without loading the entire dataset into memory. The result: fast access, even for multi-gigabyte datasets.

4. Interoperability

HDF5 is a standard format recognized by many tools and languages (C, Fortran, MATLAB, R, Julia, etc.). You can share your file and open it elsewhere without complex conversion.

5. Scalability

Whether your data are a few megabytes or multiple terabytes, HDF5 scales. Performance remains high for both data access and updates, thanks to its internal indexing and caching mechanisms.